Jean Carlo Jesús Vallejos Pongo

Chief Architect

Chief Data&AI Architect with a proven track record in building and scaling data practices across cloud ecosystems including AWS, Snowflake, Databricks, Azure, and GCP. Specialized in modern data architectures [Lakehouse, Data Mesh, MLOps/LLMOps] and Generative AI solutions powered by state-of-the-art foundation models. I believe Data and AI are not just technologies, but catalysts for cultural and organizational transformation reshaping how companies work, innovate, and make decisions.

As host of Jotatech Podcast, I explore the intersection of human creativity, data, and technology, sharing insights on how innovation and ethics can coexist to shape a better digital future.

My purpose is to turn data into business value, foster innovation, and lead strategies that embrace change, fuel creativity, and shape the future with purpose and impact.

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Experience

Jan 2026 - Present

Chief AI Architect

NTT Data Barcelona, Spain

- Lead end-to-end architecture for Data, Analytics, and GenAI initiatives in a banking environment, ensuring security, compliance, scalability, and cost efficiency.
- Define reference architectures, standards, and reusable blueprints to accelerate PoCs and drive a smooth path to production-grade delivery.
- Architect conversational AI and agentic workflows (RAG/SQL, tool-use, orchestration), integrating enterprise data sources and services.
- Design solutions on AWS with a focus on lakehouse / data mesh, data governance, and operational excellence (observability + FinOps).
- Serve as an internal instructor in the company's Data Architecture training program, delivering enablement sessions and mentoring consultants.

Apr 2025 - Oct 2025

Head of Data&AI Practice

Capitole Barcelona, Spain

- Founded and led the Data & AI Practice, scaling to 450+ consultants across Iberia (115 in Spain). Defined the strategic roadmap, operating model (Practice vs. Factory), and value proposition across Data Engineering, BI, Data Governance, Data Science/LLMs, and RPA.
- Directed and mentored cross-functional teams, fostering a culture of technical excellence, innovation, and continuous learning through tailored career paths, certifications, and knowledge-sharing programs.
- Designed and executed the Data & AI capability framework, enabling consistent delivery standards and measurable performance outcomes aligned with business objectives.
- Developed a 100% serverless AI solution to streamline practice operations and accelerate internal growth on AWS Cloud, leveraging AWS Bedrock, AWS Lambda, and Claude Sonnet models.
- Led enterprise RFPs and technical proposals on AWS, Databricks, and Azure, integrating technologies like Glue, Kinesis, Redshift Serverless, Unity Catalog, Delta Lake, Data Factory, and Fabric.
- Acted as a strategic and technical thought leader, representing the firm through keynotes, podcasts, and industry events, strengthening brand positioning and market influence.
- Managed strategic alliances with AWS and Snowflake, launching AI innovation programs, data governance frameworks, and LLM agent initiatives to drive long-term differentiation.

Mar 2022 - Present

Associate Professor

San Martin de Porres University Lima, Peru

- Teach undergraduate and postgraduate courses in Big Data, Business Intelligence, and Data Science, integrating real-world use cases, cloud ecosystems, and modern architectures (Data Lakehouse, Data Mesh, Fabric).
- Design and continuously update 16-week course programs focused on hands-on learning, covering data modeling, analytics, visualization, and AI-driven decision-making.
- Mentor students and supervise thesis projects related to Data Engineering, AI applications, and digital transformation, fostering analytical thinking and ethical data practices.
- Serve as thesis evaluator and academic advisor, guiding research on topics such as predictive analytics, data governance, and business optimization through AI.
- Promote interdisciplinary collaboration, connecting data science education with business, sustainability, and innovation.

Mar 2025 - Present

Professor Part-time

Universitat Politècnica de Catalunya Barcelona, Spain

- Designed and taught the Artificial Intelligence module of the Digital Sustainability Postgraduate Program, connecting AI methodologies with sustainable architecture and smart city applications.
- Developed a hands-on LLM project integrating IFC (Building Information Modeling) files and regulatory frameworks, enabling students to query building models and norms through a conversational AI interface.
- Introduced applied AI tools such as OpenAI APIs, RAG architectures, vector databases, semantic search, prompt engineering, fostering practical understanding of AI-driven knowledge systems in built environments.
- Mentored interdisciplinary teams of architects, engineers, and technologists, guiding them in prompt engineering, data governance, and ethical use of AI within sustainable design contexts.

Feb 2025 - Present

Professor Part-time

Universitat de Barcelona Barcelona, Spain

- Teach and design academic content on Artificial Intelligence, Big Data, and Blockchain applied to the sports industry, exploring how emerging technologies enhance performance, governance, and fan engagement.
- Develop asynchronous and synchronous learning modules covering Web3, data infrastructure [centralized vs. decentralized], AI ethics, and metaverse applications for sport organizations.
- Guide students in practical case studies and video lectures, linking data-driven decision-making with digital transformation in sports.
- Promote interdisciplinary thinking between technology, management, and athletic performance to prepare professionals for the future of AI-enabled sports ecosystems.
- Contribute to program innovation, ensuring content remains aligned with global trends in SportsTech, Data Analytics, and Digital Transformation.

Dec 2022 - Apr 2025

Data&AI Lead

Kyndryl Barcelona, Spain

- Led the design and implementation of data architectures on AWS and GCP, building Data Lake and Lakehouse frameworks to support analytics, ML, and governance use cases.
- Architected and delivered data ingestion and transformation pipelines on AWS using S3, IoT Core, DMS, Kinesis Data Streams & Firehose, Glue, Athena, Redshift, Step Functions, Lambda, and SNS, ensuring scalability and resilience.
- Designed end-to-end ML/AI architecture blueprints leveraging S3, Lambda, Rekognition, Kinesis Video Streams, SageMaker, and MLflow, enabling real-time analytics and computer vision workflows.
- Led PoC initiatives in Generative AI and Data Engineering, demonstrating business value through applied innovation and measurable performance outcomes.
- Directed a Data Lake migration from on-premise to GCP, integrating Composer, Cloud Storage, Dataform, BigQuery, and Dataproc to modernize infrastructure and optimize costs.
- Mentored junior engineers and analysts, fostering upskilling in Apache Spark, AWS Cloud, and Hadoop ecosystems.
- Conducted training sessions for non-technical users on data visualization, storytelling, and BI best practices, promoting data literacy across teams.

Nov 2021 - Dec 2022

Data Lead

Delosi Lima, Perú

- Created and led the company's first Data Analytics Center of Excellence (CoE), transforming a traditional Business Intelligence unit into a modern data-driven organization focused on advanced analytics, governance, and AI adoption.
- Led the Data Engineering team under Agile methodologies, driving the company's data modernization strategy in collaboration with the executive committee.
- Designed, implemented, and optimized end-to-end data pipelines on AWS, integrating DMS, Lambda, EventBridge, S3, Glue, EMR, Redshift, Athena, and DynamoDB to ensure scalability, automation, and data quality.
- Built and maintained the enterprise Customer Data Lake, enabling unified analytics, personalization models, and advanced reporting across business units.
- Developed and deployed ML-based analytics for demand forecasting and customer segmentation, leveraging SageMaker and EMR for scalable training and inference.
- Managed the QlikSense platform, delivering interactive HR and Financial dashboards that enhanced strategic decision-making.
- Designed and maintained a Kafka-based architecture for real-time ingestion from the MACROS transactional system, enabling near-real-time analytics.
- Planned and supervised the migration of on-premise workloads to AWS Cloud, ensuring security, cost efficiency, and business continuity.
- Presented innovative approaches to data management, AI integration, and digital transformation in executive meetings and internal workshops.

Apr 2021 - Nov 2021

Data architect

Globant Lima, Perú

- Key contributor to Johnson & Johnson's Advanced Analytics initiative for fraud detection in the global vaccine supply chain, ensuring data integrity and traceability across multiple systems.
- Designed and optimized the project's data architecture using AWS Glue, Lambda, DMS, Kinesis Data Streams, and S3, enabling automated and real-time data ingestion and processing.
- Performed advanced data analysis with AWS Athena (SQL) and Python, supporting anomaly detection and predictive modeling pipelines.
- Contributed to ENGIE Mexico's Advanced Analytics project, leading data discovery and pipeline automation design to improve operational efficiency.
- Led the migration from Denodo to AWS, architecting a scalable data platform based on S3, Lambda, and SageMaker for machine learning experimentation and deployment.

Jan 2020 - Apr 2021

Big data Engineer Lead

Indra Lima, Perú

- Key contributor to BBVA's Datio initiative, the bank's enterprise data platform for analytics and governance.
- Performed data source analysis across core banking systems (HOST) to the data governance phase, ensuring data integrity, lineage, and compliance with internal standards.
- Developed and optimized ETL processes using PowerCenter and Teradata SQL, automating ingestion workflows and improving performance.
- Created and maintained data dictionaries for Raw (Avro) and Master (Parquet) layers to enhance traceability and schema standardization.
- Executed workflow orchestration with Control-M, coordinating dependencies and ensuring SLA-compliance across ingestion pipelines.
- Administered the data governance tool to validate datasets before ingestion into the Datio platform.
- Ingested and validated data from multiple banking systems, versioning code through Bitbucket and enforcing quality checks via Hamurabi rules.
- Performed peer code reviews and processed large-scale data in HDFS using PySpark and SparkSQL, supporting analytical model readiness and downstream reporting

Jun 2019 - Dec 2019

Big data engineer

SEAT Barcelona, Spain

- Led sentiment analysis initiatives across social media and SEAT's digital platforms, providing actionable insights to the Quality Department for product improvement and customer satisfaction.
- Directed business analysis and development of AI-driven Chatbot solutions between factory operations and customers using the Inbenta platform, enhancing digital interaction and service efficiency.
- Developed new data flow channels for the Car Connected initiative, supporting real-time telemetry and connected vehicle analytics.
- Proposed and implemented Big Data projects under the Smart Launch program, leveraging Hadoop ecosystem tools (Sqoop, Storm, Kafka, Hive, HBase, Spark) for large-scale data processing.
- Designed Big Data architecture for the Car Connected project, integrating API Gateway, Kinesis, Athena, DynamoDB, and Glue to enable scalable ingestion and analytics.
- Contributed to data governance assessments within the Smart Launch Big Data project, ensuring compliance, quality, and security standards across data pipelines.

Jan 2018 - Dec 2019

PowerBI Data Analyst & Coach

Ourlines Lima, Peru

- Trained users in Microsoft Power BI, oering beginner and intermediate level classes to both IT and non-IT professionals.
- Covered foundational database concepts and data modeling.
- ETL functions with M language and DAX for table relationships.
- Provided instruction on visualization theory and KPI's for report creation.
- Used the Power BI Cloud Service for real-time data visualization and reporting.
- Developed and maintained data marts (DM) and data warehouses (DW) in SQL Server for clients in the telecommunications sector.

Jan 2017 - Jul 2018

Digital Solution Analyst

ENGIE Lima, Peru

- Managed IT resources and conducted functional analysis for digital transformation projects across ENGIE's power plants, ensuring alignment between technical solutions and business needs.
- Performed IT service audits for COSAPI Data, producing interactive Power BI dashboards to track compliance and service performance.
- Oversaw IT asset management and resource allocation using SAP ERP, improving visibility and control over infrastructure resources.
- Led the company's technology renewal initiative, coordinating upgrades of key systems and hardware across departments.
- Acted as IT Lead for the company's Business Intelligence Marathon, promoting innovation and data literacy among internal teams.
- Served as functional analyst for internal Azure-based applications, contributing to process automation and operational efficiency.

Education

Nov 2024 - Present

Executive MBA

ESADE Business School Barcelona, Spain

Sep 2025 - Oct 2025

Executive MBA – International Immersion Week

American University in Dubai Dubai, UAE

Oct 2025 - Nov 2025

Leading Change and Transformation

Aalto University Helsinki, Finland

Feb 2026 - Feb 2026

Executive MBA - International Immersion Week

Globis Business School Tokyo, Japan

Oct 2020 - Jul 2021

Master Degree in Artificial Intelligence

UCAM Murcia, Spain

Jan 2019 - Dec 2020

Postgraduate in Agile Project Management with Scrum , Kamban , Lean and xp , Project Management

IEBS Business School Barcelona, Spain

Oct 2018 - Jul 2019

Master Degree in Big Data

Universidad de La Salle Barcelona, Spain

Jan 2017 - Dec 2017

Business analytics and Big data , Data Science

National University of Engineering Lima, Peru

Mar 2011 - Nov 2017

Computing Systems Engineering

San Martin de Porres University Lima, Peru

Languages

English
Advanced

Castellano
Native

Skills

Teamwork
Public speaking
Problem-Solving
Handling conflict
Customer service
Innovation & Strategy
Leadership
Mentoring & Coaching
Public Speaking
Stakeholder Management

Certificates & Courses

AWS Certified Data Analytics Specialty

Business Intelligence Foundation Professional Certification - BIFPC